

The `slChannel` algorithm: semi-Lagrangian DNS of turbulent channel flow

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Abstract

This document is the theory manual of `slChannel`. It records, substep by substep, the discrete equations the released solver computes, together with the notation and the published results the scheme rests on. Section 1 fixes the setting and the wall-unit conventions; Section 2 collects what the semi-Lagrangian Navier–Stokes literature establishes, with the governing equations and the design decisions it forces; Section 3 specifies the implemented scheme, and §3.2 within it writes out one full time step exactly as coded in `solver.py` and `semilag.py`. The manual is versioned with the code so that the two cannot drift apart. Validation results and the analysis of the scheme’s error mechanisms are reported separately.

1 Setting and notation

Incompressible Navier–Stokes in Lagrangian form,

$$\frac{d\mathbf{u}}{dt} = -\nabla p + \nu \nabla^2 \mathbf{u} + \mathbf{f}, \quad \nabla \cdot \mathbf{u} = 0, \quad \frac{d\mathbf{x}}{dt} = \mathbf{u}(\mathbf{x}, t). \quad (1)$$

A semi-Lagrangian scheme advances the solution on a *fixed* grid by tracing, each step, the characteristic arriving at every grid node $\mathbf{x}_a$ backwards to its departure point (“foot”) $\mathbf{x}_d$, and interpolating the old field there. Wall units use u_τ and ν : $\Delta t^+ = \Delta t u_\tau^2 / \nu$, $\Delta x^+ = \Delta x u_\tau / \nu$. The `slChannel` operating points are the open channel at $\text{Re}_\tau \approx 182$ ($256^2 \times 100$, $\Delta x^+ = 7.7$) and the closed KMM [10] channel at $\text{Re}_b = 2792.8$ ($\text{Re}_\tau = 178.12$; $192^2 \times 256$ with $\Delta x^+ = 11.8$ and $256^2 \times 256$ with $\Delta x^+ = 8.8$). The Eulerian reference scheme (identical spatial discretization, AB2/CN IMEX) is capped at $\text{CFL} \approx 0.28$, i.e. $\Delta t^+ \approx 0.06$; the SL schemes run at $\Delta t^+ = 0.25$ – 0.40 .

2 What the literature proves

2.1 The premise: temporal over-resolution of Eulerian DNS

Xiu & Karniadakis [2] formalize why explicit advection wastes resolution in *time*. With $N \approx \text{Re}^{3/4}$ nodes per direction to resolve the Kolmogorov length, an eigenvalue scaling N^α of the discrete advection operator gives

$$\frac{\Delta t_{\text{CFL}}}{\tau_\eta} \propto \text{Re}^{1/2} \frac{1}{N^\alpha} \propto \text{Re}^{1/2-3\alpha/4}, \quad (2)$$

with $\alpha = 1$ (Fourier) to 2 (Chebyshev). At $\text{Re} = 10^4$ the CFL step sits one to four orders of magnitude *below* the Kolmogorov time: the smallest spatial scales are marginally resolved while the temporal scales are over-resolved by orders of magnitude. This is exactly the `slChannel` premise; at our $\text{Re}_\tau = 180$ operating point the physics limit $\Delta t^+ \approx 0.3$ – 0.4 (a fraction of the near-wall Kolmogorov time $\tau_\eta^+ \approx 2$ – 3 , and of the sweeping-decorrelation time of the smallest resolved eddies) sits 5–7 $\times$ above the Eulerian cap.

2.2 The error structure of interpolated SL advection

The rigorous convergence analysis of Falcone & Ferretti [5] (anticipated by McDonald and the meteorological literature [7]) gives, for backward integration of order k and interpolation of polynomial order P ,

$$\frac{u^{n+1} - u_d^n}{\Delta t} = \frac{du}{dt} + \mathcal{O}\left(\Delta t^k + \frac{\Delta x^{P+1}}{\Delta t}\right). \quad (3)$$

Three consequences are proved and repeatedly verified in [2, 3, 4]:

1. **Non-monotonicity in Δt .** The error *decreases* as the step grows while the interpolation term dominates (fewer remaps per unit time), reaches a minimum, then grows as Δt^k . Xiu & Karniadakis map this competition explicitly (their Figs. 4–5: at intermediate spatial resolution the minimum sits at $\Delta t \approx 0.024$ for their cone problem; at high resolution pure Δt^2 growth remains).
2. **Small- Δt over-dissipation.** The $\Delta x^{P+1}/\Delta t$ term means an SL run at *small* step accumulates interpolation error at rate $1/\Delta t$ — Boukir et al.’s Remark 7(ii) [1]. SL should be run at large Δt or not at all.
3. **Low-order interpolation is unusable.** SL with linear interpolation is algebraically equivalent to first-order upwinding (excessively dissipative); cubic is the classical minimum [7]; Giraldo showed that for spectral-element interpolation of order $P \geq 4$ dispersion and dissipation errors essentially vanish at any Courant number [6].

Stability is subtle: L^∞ -stability holds for monotone (low-order) interpolation; for order > 2 on *equidistant* stencils von Neumann analysis shows instabilities can arise unless “sliding” stencils centred on the departure point are used (what everyone, including `slChannel`, does); for nonuniform grids “ L^2 stability holds although a rigorous theory is incomplete” [4]. Two further practical constraints from [4]: trajectories must not cross,

$$\Delta t < \|J_{\mathbf{u}}^{-1}\|, \quad J_{\mathbf{u}} = \partial \mathbf{u} / \partial \mathbf{x} \quad (4)$$

(in wall units at $\text{Re}_\tau = 180$ the near-wall shear gives $\Delta t^+ \lesssim \mathcal{O}(1)$, comfortably satisfied); and boundary feet “may search for points outside the domain due to the large time step” — the proposed remedy in the literature is subcycling grid points near the boundary with $\delta t < \Delta t$, acknowledged to complicate the implementation.

2.3 Second-order schemes: the Crank–Nicolson trap and the stiffly-stable cure

This is the core methodological content of [2] and [1], and it maps one-to-one onto the `slChannel` time schemes.

Backward integration. Both papers trace feet with the explicit second-order midpoint rule; writing $\alpha = \mathbf{x}_a - \mathbf{x}_d$,

$$\alpha = \Delta t \mathbf{u}\left(\mathbf{x}_a - \frac{\Delta t}{2} \mathbf{u}(\mathbf{x}_a, t^n), t^{n+\frac{1}{2}}\right), \quad (5)$$

iterated to convergence (the implicit midpoint variant gives “almost identical results”; the explicit form is preferred for velocity fields known only numerically [2]). `slChannel`’s iterated-midpoint departure solver is exactly this.

First order. Backward-Euler characteristics, $(u^{n+1} - u_d^n)/\Delta t = (-\nabla p + \nu \nabla^2 u)^{n+1}$: unconditionally stable, robust, first order — the direct analogue of `slChannel v1`.

The Crank–Nicolson trap. The obvious second-order scheme,

$$\frac{\mathbf{u}^{n+1} - \mathbf{u}_d^n}{\Delta t} = -\nabla p^{n+1} + \nu \nabla^2 \left(\frac{\mathbf{u}^{n+1} + \mathbf{u}_d^n}{2} \right), \quad \frac{d\mathbf{x}}{dt} = \mathbf{u}^{n+\frac{1}{2}} = \frac{3}{2}\mathbf{u}^n - \frac{1}{2}\mathbf{u}^{n-1}, \quad (6)$$

“seems to be second-order in time ... however, our numerical experiments show that it is in fact still first-order” [2]: integrating (1) along the characteristic, the midpoint rule requires the *whole* right-hand side averaged between the departure point and the arrival point,

$$\frac{\mathbf{u}^{n+1} - \mathbf{u}_d^n}{\Delta t} = \frac{1}{2} \left[(-\nabla p + \nu \nabla^2 \mathbf{u})_d^n + (-\nabla p + \nu \nabla^2 \mathbf{u})^{n+1} \right]. \quad (7)$$

Equation (7) is `slChannel`’s `v2` (explicit terms and the extrapolated pressure gradient averaged between departure and arrival, trajectory-CN diffusion). The paper then reports that the three-step splitting of (7) “develops a long-term instability ... due to the explicit part of the pressure term ... it could be treated by additional projections” — documented on the Taylor vortex at $\text{Re} = 10^6$ (their Figs. 9, 16), where the CN-type scheme drifts over hundreds of time units while the stiffly-stable scheme tracks the Eulerian reference. Independently, `slChannel` hit the same wall twice: the `pc` trajectory predictor diverged until the predictor was *projected* (one extra Poisson solve — literally the “additional projection”), and the `v2` incremental-pressure bookkeeping had to enter *before* the implicit solve to avoid degenerating to the non-incremental scheme.

The stiffly-stable (BDF2) scheme. Both [1] and [2] arrive at the same cure — replace the neutrally-stable trapezoidal combination by a backward-differentiation formula along characteristics:

$$\boxed{\frac{\frac{3}{2}\mathbf{u}^{n+1} - 2\mathbf{u}_d^n + \frac{1}{2}\mathbf{u}_d^{n-1}}{\Delta t} = (-\nabla p + \nu \nabla^2 \mathbf{u})^{n+1}}, \quad (8)$$

with *two independent feet*: $\mathbf{x}_d^n$ traced over one step Δt and $\mathbf{x}_d^{n-1}$ traced over two steps $2\Delta t$, both starting from the arrival node. Boukir et al. freeze one trajectory field $U^* = 2\mathbf{u}^n - \mathbf{u}^{n-1}$ for both feet and prove (Remark 4(i)) that *continuing the far foot from the near foot* (“bending”) drops the global order to one — the feet must be independent solutions of the same backward ODE. Xiu & Karniadakis trace the near foot with the extrapolated field $\mathbf{u}^{n+1/2} = \frac{3}{2}\mathbf{u}^n - \frac{1}{2}\mathbf{u}^{n-1}$ and the far foot with $\mathbf{u}^n$; either choice preserves second order (Taylor analysis). The three-step splitting used in [2, 3],

$$\frac{\hat{\mathbf{u}} - 2\mathbf{u}_d^n + \frac{1}{2}\mathbf{u}_d^{n-1}}{\Delta t} = 0, \quad (9)$$

$$\frac{\hat{\mathbf{u}} - \hat{\mathbf{u}}}{\Delta t} = -\nabla p^{n+1}, \quad \nabla^2 p^{n+1} = \frac{1}{\Delta t} \nabla \cdot \hat{\mathbf{u}}, \quad \frac{\partial p}{\partial n} = -\nu \mathbf{n} \cdot [\hat{\mathbf{u}}/\Delta t + \nabla \times \boldsymbol{\omega}^{n+1}], \quad (10)$$

$$\frac{\frac{3}{2}\mathbf{u}^{n+1} - \hat{\mathbf{u}}}{\Delta t} = \nu \nabla^2 \mathbf{u}^{n+1}, \quad (11)$$

is non-incremental in the pressure with the rotational (Karniadakis–Israeli–Orszag [8]) boundary condition; the implicit solve and projection act with the effective step $\Delta t_{\text{eff}} = \frac{2}{3}\Delta t$. Boukir et al. prove stability under a non-restrictive step condition and error estimates of the type

$$\|\mathbf{u} - \mathbf{u}_h\| \leq C \left(\Delta t^2 + h^m + \frac{h^{m+1}}{\Delta t} \right) \quad (12)$$

for degree- m elements — second order in time, with the same interpolation-accumulation term as (3). `slChannel`’s `bdf2` scheme is the Boukir variant (frozen U^* , two independent feet, $\Delta t_{\text{eff}} = 2\Delta t/3$, non-incremental or incremental pressure), verified at self-convergence ratios 4.00 (incremental) and 2.0 (non-incremental, projection-splitting limited).

2.4 Strong vs. auxiliary form; interpolation choices

Xiu, Sherwin, Dong & Karniadakis [4] compare the *strong* form (backward particle tracking, as above) against the *auxiliary* form — solving the pure advection problem

$$\frac{\partial \tilde{\phi}}{\partial \tau} + \mathbf{u} \cdot \nabla \tilde{\phi} = 0, \quad \tilde{\phi}(\mathbf{x}, t^n) = \phi(\mathbf{x}, t^n), \quad \tau \in [t^n, t^{n+1}] \quad (13)$$

with an Eulerian substepper at its own CFL limit (the OIFS scheme of Maday, Patera & Rønquist [9]), so that $\phi_d^n = \tilde{\phi}(\mathbf{x}_d, t^{n+1})$ without any interpolation. Verdict: at intermediate steps the auxiliary form is preferable (no interpolation error, standard parallelization), but its cost grows with the number of substeps and it is *not* CFL-free; **at large time steps only the strong form is stable and its tracking cost is essentially independent of Δt** . `slChannel` is a strong-form code; this is the right branch for $\Delta t^+ \gtrsim 0.25$.

For the channel geometry specifically ([4], §3.3: Fourier $\times$ Fourier $\times$ spectral-element, i.e. structurally our grid), they reject global Fourier interpolation as “very high” cost and use *local* interpolation in the homogeneous directions — exactly the tensor-product-stencil choice of `slChannel` — comparing Lagrange stencils of orders 3–7 against *cubic Hermite* with the nodal derivatives computed spectrally (FFT differentiation): third-order Hermite matches seventh-order Lagrange accuracy at about half the cost, and retains exponential convergence where fixed-order Lagrange saturates. This is the closest published relative of `slChannel`’s prefiltered-spline `field_interp` path (C^2 B-spline in x, y via FFT symbol division + Hermite evaluation in z).

2.5 The one turbulence result: Xu, Xiu & Karniadakis (2002)

Reference [3] is the only published SL turbulent channel DNS in this line, and its parameters land startlingly close to ours:

	Xu et al. 2002 [3]	<code>slChannel</code> (2026)
Scheme	BDF2 characteristics, Eqs. (8)–(11)	same skeleton (<code>bdf2</code> , Boukir feet)
Discretization	Fourier $\times$ SE(Jacobi) $\times$ Fourier, 48^3	FD2 staggered, $192^2\text{--}768^2 \times 100\text{--}256$
Interpolation	<i>global</i> modal (all Fourier modes)	local tricubic/triquintic, Triton
Re_τ	116	178–182 (exact KMM Re_b)
Δt^+	0.2645 ($10\times$ Eulerian Δt)	0.25–0.40 ($4\text{--}7\times$)
Statistics window	$T = 20$ convective units	70 washouts (≈ 880 c.u., $10^4 t^+$)
Validation	Eulerian twin only, mean + rms	Eulerian twin + KMM/MKM data, profiles, stresses, spectra
Cost	$10\times/\text{step} \Rightarrow \text{net } 1.0\times$	$1.3\text{--}2.1\times/\text{step} \Rightarrow \text{net } 2\text{--}3.2\times$

Their findings: (i) the SL run is *stable at marginal resolution without de-aliasing* where the Eulerian twin is unstable at any Δt — interpolation supplies the missing high- k dissipation; (ii) mean velocity indistinguishable from the Eulerian reference “despite the very large time step”; (iii) *rms intensities differ, most visibly an elevated streamwise peak*, which they attribute to time-averaging error by comparing $T = 5$ vs $T = 20$ Eulerian averages; (iv) the global interpolation made the method $10\times$ slower per step, cancelling the $10\times$ step gain — they close by calling for “better local interpolation procedures” and report no follow-up channel study. Two details deserve emphasis. First, their $\Delta t^+ = 0.2645$ sits *just below* the two-foot stability boundary $\Delta t^+ \approx 0.30$ that `slChannel` mapped in 2026 — they were one courageous step-doubling away from a mysterious blow-up. Second, their elevated streamwise rms is the same signature `slChannel` measures with $44\times$ longer averaging ($+5.6\%$ peak u'_{rms}^+ at $\Delta x^+ = 11.8$); with converged statistics it is demonstrably *not* an averaging artifact.

3 The slChannel scheme

3.1 Scheme summary

Staggered MAC finite differences (x, y uniform periodic, z tanh-stretched), `torch.float64` infrastructure with fp32-accumulate Triton gather kernels for the interpolation. Feet by iterated midpoint (5) with trilinear (default) or tricubic trajectory sampling; fields remapped by tensor-product Lagrange stencils (tricubic $P=3$ / triquintic $P=5$; nonuniform-node weights in z against the actual grid, analytic inverse tanh map for stencil location).

The time scheme is Boukir BDF2 (8) and nothing else: frozen $U^* = 2\mathbf{u}^n - \mathbf{u}^{n-1}$, two independent feet at Δt and $2\Delta t$, update $\hat{u} = (4\bar{u} - \bar{\bar{u}})/3$, $\theta=1$ z -solve and FFT projection at $\Delta t_{\text{eff}} = 2\Delta t/3$, with `noninc` (production) or `inc` pressure — velocity self-convergence ratios 2.0 and 4.00 respectively. Mass flux is imposed exactly by a uniform divergence-free shift (CaNS convention); Reynolds stresses are accumulated about the *time* mean at each component’s own staggered nodes.

The development campaigns also exercised a first-order backward-Euler splitting (`v1`), a departure-averaged Crank–Nicolson scheme with incremental pressure (`v2`, self-convergence 3.93), projected predictor–corrector trajectories (`pc`, 3.78), and a C^2 prefiltered spline remap. All are reported here as *evidence*, and all were removed from the code once BDF2 won the stability-versus-accuracy trade: the released solver implements the single scheme above. An Eulerian IMEX path is retained purely as a like-for-like reference for validation and benchmarking.

3.2 One step in full: the discrete equations

This subsection writes out every substep as implemented in `solver.py/semilag.py`, in execution order. Notation: $V^n = (u, v, w)^n$ is the staggered MAC velocity (u at x -faces, v at y -faces, w at z -faces, one ghost layer per side; pressure at cell centres); G_h and D_h are the staggered gradient and divergence; L_{xy} and L_z the horizontal and wall-normal discrete Laplacians (second-order central, L_z on the stretched metric); $\mathcal{A}[\phi](\mathbf{x}_d)$ denotes the interpolation of a field ϕ at the departure points of the component grid on which ϕ lives. All ghost layers are rebuilt (*BC application*) after every operation that leaves them stale.

(1) Trajectory velocity. A single advecting field $\mathbf{U}^*$, defined per scheme:

$$\mathbf{U}^* = \begin{cases} \frac{3}{2}V^n - \frac{1}{2}V^{n-1} & \text{v1/v2, ab2 (target } t^{n+1/2}), \\ V^n & \text{traj_extrapolation: none (diagnostic, } \mathcal{O}(\Delta t)), \\ \frac{1}{2}(V^n + \tilde{V}^{n+1}) & \text{pc (projected predictor } \tilde{V}^{n+1}, \text{ see below),} \\ 2V^n - V^{n-1} & \text{bdf2 (frozen for both feet, target } t^{n+1}). \end{cases} \quad (14)$$

For `pc`, the predictor is a complete first-order step (SL advection of V^n by itself, plus the horizontal viscous tendency, plus the *lagged* pressure gradient $-G_h p^{n-1/2}$, then a backward-Euler z -solve) which is then *projected* — one extra Poisson solve. The projection is not optional: an unprojected predictor carries an $\mathcal{O}(\Delta t)$ divergence, SL trajectories through a divergent mid-velocity are compressive, and the scheme blows up ($u_\tau + 127\%$).

(2) Departure points: iterated midpoint. For every arrival node $\mathbf{x}_a$ of each component grid, with $K = 2$ iterations (`n_traj_iters`):

$$\mathbf{x}_m^{(0)} = \mathbf{x}_a, \quad \mathbf{x}_m^{(s)} = \mathbf{x}_a - \frac{\Delta t}{2} \mathbf{U}^*(\mathbf{x}_m^{(s-1)}), \quad s = 1, \dots, K, \quad \mathbf{x}_d = 2\mathbf{x}_m^{(K)} - \mathbf{x}_a, \quad (15)$$

i.e. $\mathbf{x}_d = \mathbf{x}_a - \Delta t \mathbf{U}^*(\mathbf{x}_m^{(K)})$: the explicit iterated midpoint rule (5). $\mathbf{U}^*$ is sampled at the (staggered) midpoints by trilinear interpolation on the fast path (`traj_order: 2`; its C^0 interpolant

caps *self*-convergence at fixed grid at $\mathcal{O}(\Delta t)$ with a small h^2 coefficient — tricubic sampling, `traj_order`: 4, restores clean $\mathcal{O}(\Delta t^2)$, ratio 3.93). x, y are unwrapped modulo the period; z is clamped to the wall-adjacent node range and every clamp is counted (`clamped` diagnostic; measured 0 over $10^4 t^+$ in the closed channel). For `bdf2` the same $\mathbf{U}^*$ traces *two independent* feet from the same arrival node, (15) with Δt and with $2\Delta t$ — never the far foot continued from the near one (Boukir Remark 4(i)).

(3) Remap. The old field is evaluated at the feet by a tensor-product Lagrange stencil of $P+1$ nodes per direction ($P = 3$ tricubic, $P = 5$ triquintic):

$$\mathcal{A}[\phi](\mathbf{x}_d) = \sum_{i,j,k} w_i^x(x_d) w_j^y(y_d) w_k^z(z_d) \phi_{ijk}^n, \quad w_i(\xi) = \prod_{m \neq i} \frac{\xi - \xi_m}{\xi_i - \xi_m}. \quad (16)$$

In x, y the nodes are uniform and the weights close-form; in z the weights are built *against the actual stretched nodes* z_c/z_f (uniform- ξ weights on the tanh image would silently lose an order: z_c are face midpoints, not the tanh-map image), the stencil is located by the analytic inverse tanh map plus one node compare (no search), sliding stencils centred on the foot in the interior, one-sided at the walls. The index/weight set of (16) is computed once per component and step and reused for every field remapped on that grid — this is what makes the `v2` departure-evaluated right-hand sides in (18) nearly free. The alternative C^2 path (`field_interp`: `spline`) replaces (16) by a prefiltered cubic B-spline in x, y (FFT symbol division) and a nonuniform C^2 cubic spline in z with Hermite evaluation.

(4) Explicit horizontal tendency. On V^n , $R_{xy} = \nu L_{xy} V^n$ (no body force here — the flux constraint is substep 8). `v2` time-centres it by AB2, $R_{xy}^{n+1/2} = \frac{3}{2} R_{xy}^n - \frac{1}{2} R_{xy}^{n-1}$; `bdf2` extrapolates to t^{n+1} , $R_h = 2R_{xy}^n - R_{xy}^{n-1}$ (`bdf2_xy_rhs`: `extrap`; `lagged` uses R_{xy}^n).

(5) Advected predictor. Per scheme:

$$\text{v1: } u^* = \mathcal{A}[V^n](\mathbf{x}_d) + \Delta t R_{xy}, \quad (17)$$

$$\text{v2: } u^* = \mathcal{A}[V^n](\mathbf{x}_d) + \Delta t \left[\frac{1}{2} (\mathcal{R}_d + \mathcal{R}_a) + \frac{1}{2} (\nu L_z V^n)_d \right], \quad \mathcal{R} = R_{xy}^{n+1/2} - G_h p_{\text{ext}}, \quad (18)$$

$$\text{bdf2: } u^* = \frac{4}{3} \mathcal{A}[V^n](\mathbf{x}_d^{\Delta t}) - \frac{1}{3} \mathcal{A}[V^{n-1}](\mathbf{x}_d^{2\Delta t}) + \Delta t_{\text{eff}} [R_h - G_h p_{\text{ext}}], \quad \Delta t_{\text{eff}} = \frac{2}{3} \Delta t. \quad (19)$$

In (18) subscripts d/a mean the field evaluated at the departure point (via the reused stencil of substep 3, with BC-consistent ghosts so near-wall feet interpolate meaningful values) and at the arrival node: the *entire* explicit right-hand side, including the extrapolated pressure gradient $p_{\text{ext}} = 2p^{n-1/2} - p^{n-3/2}$, is averaged along the characteristic — this is exactly what distinguishes the true second-order scheme (7) from the first-order trap (6). The z -viscous term enters with its departure half only; its arrival half is the implicit solve below. In (19), $p_{\text{ext}} = 2p^n - p^{n-1}$ (full levels) for `bdf2_pressure`: `inc` and is omitted for `noninc`; dividing Boukir’s update (8) by the BDF2 leading coefficient $\frac{3}{2}$ is what produces the $\frac{4}{3}/\frac{1}{3}$ foot weights and the effective step Δt_{eff} carried by every subsequent substep. The pressure term must enter *here*, before the implicit solve: adding it after is algebraically identical to the non-incremental scheme (verified analytically and numerically).

(6) Implicit wall-normal diffusion. One tridiagonal solve per horizontal position and component (θ -scheme on the stretched metric, no-slip rows at the walls):

$$(I - \theta \delta t \nu L_z) u^{**} = u^* + (1 - \theta) \delta t \nu L_z u^*, \quad (\theta, \delta t) = \begin{cases} (\frac{1}{2}, \Delta t) & \text{v1 (CN at arrival),} \\ (1, \frac{\Delta t}{2}) & \text{v2 (arrival half of trajectory-CN),} \\ (1, \Delta t_{\text{eff}}) & \text{bdf2.} \end{cases} \quad (20)$$

(7) Projection. With $\delta t = \Delta t$ (v1/v2) or Δt_{eff} (bdf2),

$$L_h \phi = \frac{D_h \mathbf{u}^{**}}{\delta t}, \quad \mathbf{u}^{n+1} = \mathbf{u}^{**} - \delta t G_h \phi, \quad p^{\text{new}} = \begin{cases} \phi & \text{v1, bdf2-noninc,} \\ p_{\text{ext}} + \phi & \text{v2, bdf2-inc.} \end{cases} \quad (21)$$

The Poisson solve diagonalizes x, y by real FFTs with the second-order modified wavenumbers $\tilde{k}^2 = 2(1 - \cos k\Delta)/\Delta^2$, leaving one tridiagonal system in z per mode pair (k_x, k_y) with homogeneous Neumann ends (built into the staggered stencil — no explicit pressure BC, hence no rotational-BC machinery of (10) is needed); the singular Neumann–Neumann $(0, 0)$ mode is pinned. On the incremental branches the splitting error of (21) is $\mathcal{O}(\Delta t^2)$ (self-convergence ratios 3.93/4.00); the non-incremental branch is first-order in the projection but robust (ratio 2.0).

(8) Exact bulk-flux constraint. Last, the CaNS-convention mass flux enforcement:

$$c = U_b - \langle u^{n+1} \rangle_V, \quad u^{n+1} += c \text{ (uniformly)}, \quad f = c/\Delta t, \quad (22)$$

where $\langle \cdot \rangle_V$ is the volume average. A uniform shift is divergence-free, so it commutes with the projection just performed; the driving force f is *diagnosed* from the correction rather than steered towards it. This replaces an integral controller whose error dynamics were an undamped oscillator with $1/\Delta t$ gain — under which any Δt sweep partly measured the controller instead of the scheme.

Bootstrapping. `bdf2` assumes constant Δt (the $2\Delta t$ foot and the $\frac{4}{3}/\frac{1}{3}$ weights); on the first step, after a restart, or on any Δt change the step degenerates to one BDF1 step (backward-Euler characteristics: $\mathbf{U}^* = V^n$, single foot, $\theta = 1$ at full Δt) which seeds the V^{n-1} and R_{xy}^{n-1} histories — one $\mathcal{O}(\Delta t^2)$ local error, the same policy as the documented AB2 restart re-bootstrap. `bdf2` configs pin Δt (`dt_update_interval`: 0).

Note on tooling

Parts of the implementation, the analysis scripts and the drafting of this report were carried out with the assistance of Claude (Anthropic). All results reported here were produced by the code in the repository and verified by the author, who takes full responsibility for their content.

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